QS-VC

QUANTUM-SAFE VIDEO CONFERENCING

PRODUCT DATA SHEET

Next-Generation Video Conferencing with Post-Quantum Encryption

QS-VC is a quantum-safe video conferencing platform engineered with NIST-approved post-quantum cryptography (PQC). It delivers crystal-clear HD/4K video and audio, real-time AI-powered captions and translation, and end-to-end encrypted media streams — ensuring your calls are secure against both today's and tomorrow's quantum computing threats.

99.995%	10K+	<150ms	99+
PLATFORM SLA	PARTICIPANTS/MEETING	AUDIO LATENCY	LANGUAGES STT

CORE VC CAPABILITIES

Quantum-Safe Encryption

- CRYSTALS-Kyber-1024 + X25519 hybrid key exchange
- CRYSTALS-Dilithium-5 + Ed25519 hybrid signatures
- End-to-end encrypted media (SFrame, zero-knowledge SFU)
- AES-256-GCM with 60-second key rotation
- TLS 1.3 with PQC extensions for signaling

AI Meeting Intelligence

- Real-time speech-to-text (Whisper, 99+ languages)
- Live subtitle translation (NLLB-200, 200+ pairs)
- Voice-to-voice translation with voice cloning
- AI noise suppression (RNNoise + custom U-Net)
- Post-meeting summarization & action items (LLaMA 3)

HD/4K Video & Audio

- VP9/AV1 video codecs with Simulcast & SVC layers
- Opus 48kHz stereo audio, adaptive bitrate (GCC)
- Smart camera auto-framing (MediaPipe + YOLOv8)
- Virtual backgrounds & blur (SegFormer)
- Screen sharing with annotation & remote control

Meeting & Collaboration

- Breakout rooms, waiting lobby, raise hand, reactions
- CRDT-based real-time whiteboard (Yjs)
- Cloud & on-prem recording (MinIO/S3, MP4)
- Live streaming: RTMP → HLS/DASH via CDN
- SIP/H.323 gateway for legacy room systems

TECHNICAL SPECIFICATIONS

SPECIFICATION	DETAILS
Media Transport	SRTP/DTLS 1.3 + Quantum-TLS, VP9/AV1 + Opus 48kHz stereo
Signaling	WebSocket + JSON-RPC 2.0, <3s join time (P95)
SFU Engine	mediasoup with simulcast, SVC, cascade (multi-node mesh)
Client Platforms	Web (React 18), Desktop (Electron 28), Mobile (React Native)
AI Runtime	NVIDIA Triton + TensorRT, GPU autoscaling per workload
HA & Failover	Active-passive SFU pairs <2s failover, blue-green deploys
Bandwidth Adapt.	5-tier auto: 4K → 720p → 360p → 180p → audio-only

DEPLOYMENT MODELS

SaaS (Cloud)

Multi-tenant Kubernetes, auto-scaling, per-user subscription.

On-Premise

Air-gapped, HSM-backed, zero internet dependency.

Hybrid

Edge media nodes + cloud orchestration, 72hr offline grace.

SECURITY STANDARDS

- NIST PQC Level 5
- Zero-Knowledge SFU
- E2EE (SFrame)
- HSM Key Management
- Tamper-Proof Audit Logs
- 60s Key Rotation

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